

LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report reviews two primary patents relevant to nitrile butadiene rubber (NBR) with emphasis on low-acrylonitrile embodiments and NBR-centered processing or formulation. The selected patents span two distinct technical directions: one patent is centered on an electrophotographic electro-conductive member that explicitly uses NBR as one of the diene-based rubbers in a matrix/domain architecture, while the other patent focuses on low molecular weight nitrile rubber polymers and their use in polymer composites, including a route that starts from nitrile polymer feedstock and proceeds through metathesis and hydrogenation. Across the set, the disclosed NBR-related material definitions are broader than a dedicated low-ACN synthesis patent, but both documents provide useful compositional and processing context for NBR materials. The first patent discloses NBR in a mixed-rubber electro-conductive layer, with examples showing NBR grades of different nitrile contents, including 15%, 18.0%, 27.5%, 35%, 40.0%, and 41.5% in the materials section, and comparative examples using NBR as either domain or matrix rubber. The second patent discloses a statistical butadiene-acrylonitrile copolymer substrate with 34 mol% acrylonitrile and describes solvent-based metathesis in monochlorobenzene followed by hydrogenation to produce lower-Mooney nitrile rubber feedstocks for composites. Major process trends include solution-phase chemistry in organic solvent for molecular-weight reduction/hydrogenation, and compounding/open-mill processing for final articles. Major formulation trends include use of carbon black, zinc oxide, sulfur or peroxide curatives, and auxiliary additives to tune viscoelasticity, conductivity, and processability. Overall, the landscape indicates that low-acrylonitrile NBR is represented more through disclosed NBR grades and downstream processing/composite use than through a dedicated low-ACN polymerization recipe in the extracted evidence.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US11347156B2
- Patent Title: Electro-conductive member for electrophotography, process cartridge for ...
- Assignee: Canon Kabushiki Kaisha
- Jurisdiction: US

- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose NBR polymerization; it discloses an electro-conductive member for electrophotography in which the electro-conductive layer is built from a matrix/domain structure using diene-based rubbers. NBR is used as one of the rubber components in the matrix or domain, with the performance controlled by rubber pairing, SP-value difference, tan delta ratio, and conductive particle loading. - Nitrile content: 15% — NBR (1) material description - Nitrile content: 41.5% — NBR (2) material description - Nitrile content: 27.5% — NBR (3) material description - Nitrile content: 18.0% — NBR (4) material description - Nitrile content: 35% — NBR (5) material description - Nitrile content: 40.0% — NBR (6) material description - SP value: 17.2 (J/cm³)^{0.5} — NBR (1) material description - SP value: 20.6 (J/cm³)^{0.5} — NBR (2) material description - SP value: 18.8 (J/cm³)^{0.5} — NBR (3) material description - SP value: 17.4 (J/cm³)^{0.5} — NBR (4) material description - SP value: 20.0 (J/cm³)^{0.5} — NBR (5) material description - SP value: 20.4 (J/cm³)^{0.5} — NBR (6) material description - NBR volume fraction: 28.7 vol% — Table 5-1, Example 1 - NBR volume fraction: 29.1 vol% — Table 5-1, Example 2 - NBR volume fraction: 28.9 vol% — Table 5-1, Example 3 - NBR volume fraction: 20.6 vol% — Table 5-1, Example 4 - NBR volume fraction: 20.2 vol% — Table 5-1, Example 5 - NBR volume fraction: 30.6 vol% — Table 5-1, Example 26 - NBR volume fraction: 30.4 vol% — Table 5-1, Example 27 - NBR volume fraction: 30.7 vol% — Table 5-1, Example 28 - NBR volume fraction: 30.5 vol% — Table 5-1, Example 29 - NBR volume fraction: 19.4 vol% — Table 5-1, Example 30 - NBR volume fraction: 19.2 vol% — Table 5-1, Example 31 - NBR volume fraction: 19.3 vol% — Table 5-1, Example 32 - NBR volume fraction: 19.2 vol% — Table 5-1, Example 33 - NBR volume fraction: 11.5 vol% — Table 5-1, Example 34 - NBR volume fraction: 14.4 vol% — Table 5-1, Example 35 - NBR volume fraction: 38.2 vol% — Table 5-1, Example 36 - NBR volume fraction: 40.5 vol% — Table 5-1, Example 37 - Difference of absolute SP values: 0.2 to 4.0 (J/cm³)^{0.5} — Claims - $\tan \delta_1 / \tan \delta_2$: 0.45 to 2.00 — Claims - Electro-conductive particles in domains: 50 to 150 parts by mass per 100 parts by mass of second rubber — Claims - Reactor temperature: 23° C. — claim/evaluation condition - Relative humidity: 50% — claim/evaluation condition - Frequency: 80 Hz — claim/evaluation condition - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - acrylonitrile content: 15% to 41.5% nitrile content disclosed for specific NBR grades; 15% and 18.0% are the lowest explicitly listed values

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Not a polymerization patent; electro-conductive layer compounding and molding in a rubber-based solid-state formulation.

acrylonitrile content - 15% to 41.5% nitrile content disclosed for specific NBR grades; 15% and 18.0% are the lowest explicitly listed values

Relevant Experimental Evidence - Example 1: Demonstrated an electro-conductive member using NBR in the domain composition with a matrix/domain structure and conductive particle-filled domains. - Examples 2-39: Showed multiple rubber-pair combinations, including several NBR-containing matrix/domain systems, and correlated SP-value differences and viscoelastic ratios with structure formation. - Example 40: Demonstrated use of a selected electro-conductive member as a transfer member under high-speed printing conditions.

Technical Relevance This patent is relevant because it explicitly includes NBR as a key rubber component in a functional material system and discloses several NBR grades with low-to-moderate nitrile contents, including 15% and 18.0%. Although not a synthesis patent, it provides direct evidence of NBR material selection and performance context for low-acrylonitrile grades.

Patent 2

- Patent Number: US7262244B2
- Patent Title: Polymer composites comprising low molecular weight nitrile rubber
- Assignee: Lanxess Inc.
- Jurisdiction: US
- Publication Year: 2007
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a two-step route for low-Mooney nitrile rubber polymers: first a metathesis reaction in monochlorobenzene using a ruthenium catalyst and ethylene co-olefin, then hydrogenation in the same reactor using Wilkinson's catalyst and hydrogen. The resulting nitrile rubber polymer is then compounded with fillers and curatives to make polymer composites. - Cement concentration: 6% — Example 1 - Co-olefin: ethylene — Example 1 - Co-olefin concentration: 500 psi — Example 1 - Agitator speed: 600 rpm — Example 1 - Reactor temperature: 80° C. — Example 1 - Catalyst loading: 0.25 phr — Example 1 - Solvent: monochlorobenzene — Example 1 - Acrylonitrile content: 34 mol % — Example 1 substrate - Mooney viscosity ML (1 + 4) @ 100 deg. C.: 35 — Example 1 substrate - Polymer mass: 9 kg — Example 1 - Monochlorobenzene mass: 141 kg — Example 1 - Hydrogen pressure: 1200 psi — Example 1 - Agitator speed: 600 rpm — Example 1 hydrogenation - Reactor temperature: 138° C. — Example 1 hydrogenation - Catalyst loading (Wilkinson's): 0.075 phr — Example 1 hydrogenation - Triphenylphosphine: 1 phr — Example 1 hydrogenation - Hydrogen degassing: 3 times with H₂ (100 psi) — Example 1 - Temperature ramp: 130° C. to 138° C. — Example 1 hydrogenation - Mn: 37000 — Table 2, Polymer from Exp. 1 - Mw: 76000 — Table 2, Polymer from Exp. 1 - PDI: 2.1 — Table 2, Polymer from Exp. 1 - ML 1 + 4 @ 100° C.: 6.5 — Table 2, Polymer from Exp. 1 - Mn: 97000 — Table 2, Therban A3407 - Mw: 314000 — Table 2, Therban A3407 - PDI: 3.2 — Table 2, Therban A3407 - ML 1 + 4 @ 100° C.: 70.5 — Table 2, Therban A3407 - Carbon black: 50 parts — Table 1 - Maglite D: 3 parts — Table 1 - Naugard 445: 1 part — Table 1 - Plasthall TOTM: 5 parts — Table 1 - Vulkanox ZMB-2/C5: 0.4 parts — Table 1 -

Zinc oxide: 3 parts — Table 1 - DIAK #7: 1.5 to 2.8 parts — Table 1 - Vulcup 40KE: 7.5 to 14 parts — Table 1 - Compression set: 20.14% to 22.85% — Table 3 - Hardness Shore A: 68 to 73 — Table 3 - DIN abrasion: 76 to 126 — Table 3 - Heat rise: 24° C. to 32° C. — Table 3 - Permanent set: 0.2% to 0.7% — Table 3 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - acrylonitrile content: 34 mol % — starting statistical butadiene-acrylonitrile copolymer in Example 1

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic solvent solution polymerization/solution processing in monochlorobenzene cement.

acrylonitrile content - 34 mol % — starting statistical butadiene-acrylonitrile copolymer in Example 1

Relevant Experimental Evidence - Example 1: Demonstrated metathesis of a 34 mol% acrylonitrile butadiene copolymer in monochlorobenzene followed by hydrogenation to a low-Mooney nitrile polymer. - Examples 2-6: Showed that low-Mooney nitrile rubber composites could be compounded on an open mill with fillers and curatives while retaining useful physical properties. - Table 2: Confirmed the polymer from Example 1 had much lower Mooney viscosity and narrower PDI than the reference grade. - Table 3: Showed that the low-Mooney polymer composites maintained good tensile, hardness, compression set, and abrasion properties.

Technical Relevance This patent is highly relevant because it directly addresses nitrile rubber feedstock and processing, including a disclosed acrylonitrile content of 34 mol% in the starting copolymer. While not a low-ACN synthesis patent per se, it provides a concrete NBR-based route and a composition benchmark within the nitrile rubber family.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	acrylonitrile content	Key Findings
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US11347156B2	Canon Kabushiki Kaisha	2022	No water-related process step disclosed in the extracted evidence.	15% to 41.5% nitrile content disclosed for specific NBR grades; 15% and 18.0% are the lowest explicitly listed values	Example 1: Demonstrated an electro-conductive member using NBR in the domain composition with a matrix/domain structure
US7262244B2	Lanxess Inc.	2007	No water-related process step disclosed in the extracted evidence.	34 mol % — starting statistical butadiene-acrylonitrile copolymer in Example 1	Example 1: Demonstrated metathesis of a 34 mol% acrylonitrile butadiene copolymer in monochlorobenzene followed by hydro

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US11347156B2 is a formulation/application patent using commercial NBR grades, with explicitly listed nitrile contents spanning 15% to 41.5%; the lowest disclosed grade is 15% nitrile content (NBR (1)).
- US7262244B2 is a synthesis/process patent for nitrile rubber feedstock and discloses a starting butadiene-acrylonitrile copolymer with 34 mol% acrylonitrile, which is substantially higher than the lowest NBR grades listed in US11347156B2.
- US11347156B2 is solid-state compounding/molding of rubber compositions, whereas US7262244B2 is an organic-solvent solution process using monochlorobenzene followed by hydrogenation.

- US7262244B2 discloses explicit catalyst and reaction conditions for metathesis/hydrogenation (0.25 phr Grubb's catalyst, 0.075 phr Wilkinson's catalyst, 500 psi ethylene, 1200 psi hydrogen), while US11347156B2 focuses on rubber pairing, SP-value difference, and conductive particle loading rather than polymerization chemistry.
- Both patents use NBR as a functional material platform, but only US7262244B2 provides a direct synthetic route to modified nitrile rubber feedstock; US11347156B2 provides downstream performance context for NBR-containing materials.

4. CONCLUSION

Across the two selected primary patents, low-acrylonitrile NBR is represented more by disclosed commercial grades and downstream use than by a dedicated low-ACN polymerization recipe. US11347156B2 explicitly lists NBR grades down to 15% nitrile content and uses NBR in a matrix/domain electro-conductive member, while US7262244B2 discloses a solvent-based metathesis/hydrogenation route starting from a 34 mol% acrylonitrile butadiene copolymer to produce low-Mooney nitrile rubber feedstocks for composites. Together, they show both material selection and processing routes relevant to NBR performance, with the clearest low-ACN evidence coming from the NBR grade list in US11347156B2.

5. REFERENCES

- US11347156B2 | Electro-conductive member for electrophotography, process cartridge for ... | Canon Kabushiki Kaisha | US | 2022 | <https://patents.google.com/patent/US11347156B2/en>
- US7262244B2 | Polymer composites comprising low molecular weight nitrile rubber | Lanxess Inc. | US | 2007 | <https://patents.google.com/patent/US7262244B2/en>